

Individual Assingment 5 JM

June McNally

2026-05-10

Set Up

Packages

```
# geberal and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
```

```

# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

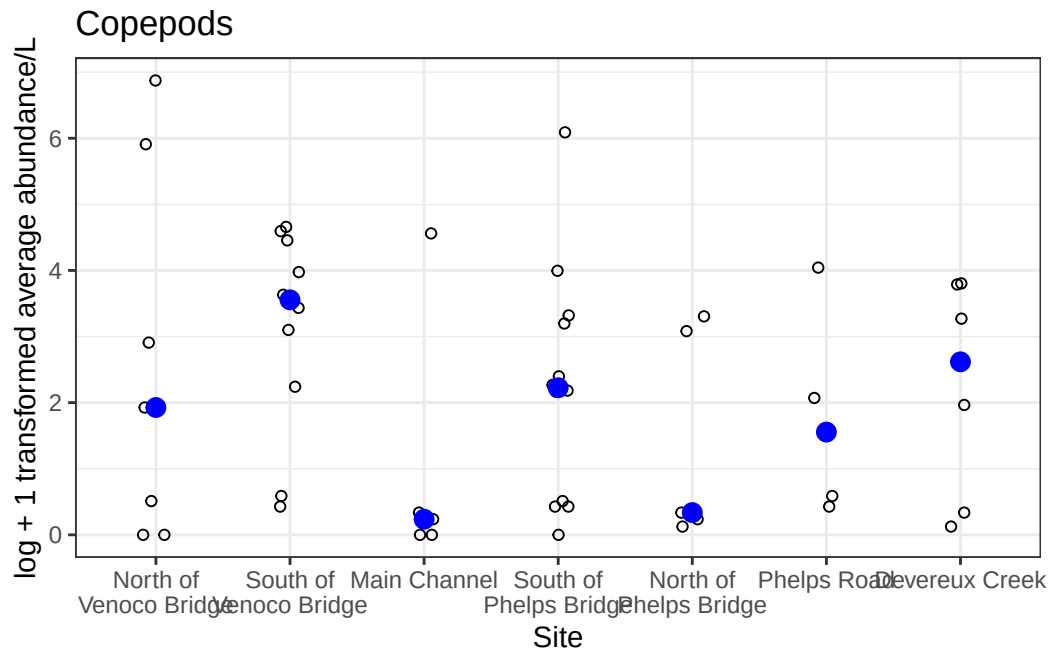
```

```
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
  site_full = fct_rev(site_full))
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))
```

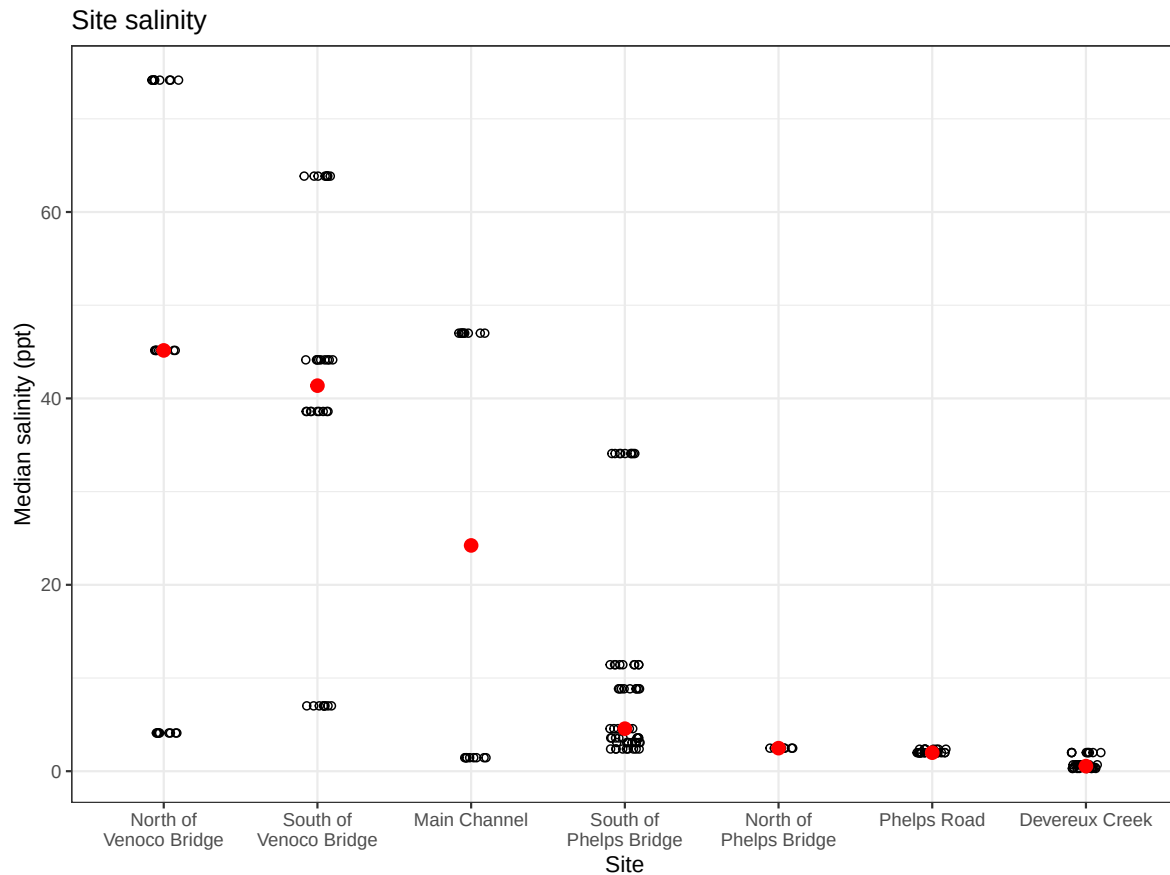
```
copepods_plot <- ggplot(data = copepods,
  mapping = aes(x = site_full, y = log_ave_lit)) +
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  stat_summary(geom = "point", fun = "median", size = 3, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site", y = "log + 1 transformed average abundance/L", title = "Copepods") +
  theme_bw()

copepods_plot
```



```
salinity <- ggplot(data = aq_ins_clean,
  mapping = aes(x = site_full,
    y = med_sal)) +
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
  scale_x_discrete(labels = label_wrap(width = 15)) +
  stat_summary(fun = median,
    color = "red",
    size = 0.5) +
  labs(x = "Site",
    y = "Median salinity (ppt)",
    title = "Site salinity") +
  theme_bw()
```

salinity



Problems

1. Visualizing differences across sites in major taxon abundance

a. Planning figure

-X axis: site (site_full) -Y axis: ostracod abundance (log + 1 transform) -geometry to represent average abundance/L: geom_jitter() -geometry to represent medians: stat_summary()

b. Wrangle data

```
ostracods <- aq_ins_clean |>
  filter(taxon == "ostracod") |>
```

```
mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87	8.32	NA	10.3
2	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
3	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
4	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA
5	2022-08-10_NPB	2022-08-10 00:00:00	NPB	ostr~	2.67	8.28	8.84	9.81
6	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87
7	2023-11-14_NPB	2023-11-14 00:00:00	NPB	ostr~	18	NA	NA	NA
8	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64
9	2022-11-19_NDC	2022-11-19 00:00:00	NDC	ostr~	6	NA	NA	NA
10	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	ostr~	0.133	7.71	1.98	9.59

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>

c. Coding figure

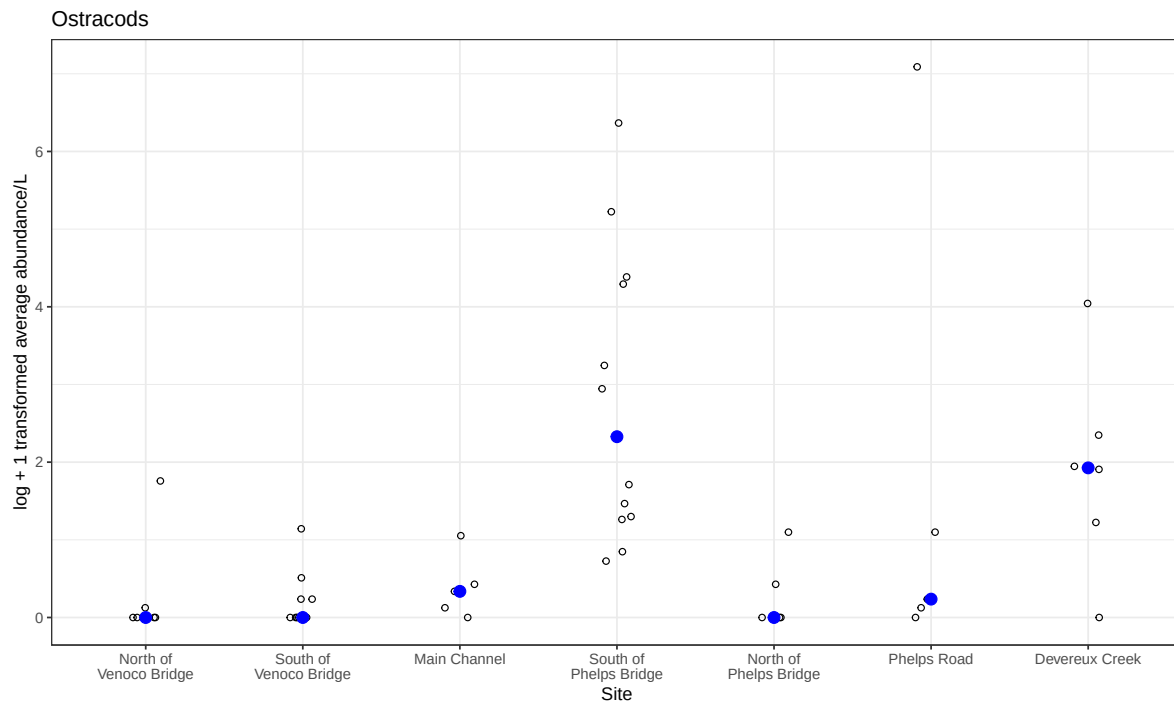
```
ostracod_plot <- ggplot(data = ostracods,
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # small open circles for individual measurements
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # blue filled point for median
  stat_summary(geom = "point",
    fun = "median",
    size = 3,
    color = "blue") +
  # wrap long site names
  scale_x_discrete(labels = label_wrap(15)) +
  # labels
  labs(x = "Site",
```

```

y = "log + 1 transformed average abundance/L",
title = "Ostracods") +
# same theme as key
theme_bw()

ostracod_plot

```



d. Adding ostracod panel

```

salinity / copepods_plot / ostracod_plot

```